

AI Powered EDA Report

Dataset Summary

```
{'Rows': 200, 'Columns': 9, 'Column Names': ['EmployeeID', 'Name', 'Age', 'Gender', 'Department', 'Experience', 'Salary', 'Performance', 'City'], 'Data Types': EmployeeID int64 Name str Age int64 Gender str Department str Experience int64 Salary int64 Performance str City str dtype: str, 'Missing Values': EmployeeID 0 Name 0 Age 0 Gender 0 Department 0 Experience 0 Salary 0 Performance 0 City 0 dtype: int64, 'Duplicate Rows': np.int64(0), 'Statistics': EmployeeID Name Age Gender Department Experience Salary Performance City count 200.000000 200 200.000000 200 200 200.000000 200.000000 200 200 unique NaN 200 NaN 2 6 NaN NaN 3 5 top NaN Employee_1 NaN Female IT NaN NaN Excellent Pune freq NaN 1 NaN 106 42 NaN NaN 74 49 mean 1100.500000 NaN 34.435000 NaN NaN 10.335000 83369.970000 NaN NaN std 57.879185 NaN 7.007231 NaN NaN 6.854758 34320.852887 NaN NaN min 1001.000000 NaN 22.000000 NaN NaN 0.000000 26501.000000 NaN NaN 25% 1050.750000 NaN 29.000000 NaN NaN 4.750000 53997.250000 NaN NaN 50% 1100.500000 NaN 35.000000 NaN NaN 11.000000 88090.500000 NaN NaN 75% 1150.250000 NaN 40.250000 NaN NaN 17.000000 114769.000000 NaN NaN max 1200.000000 NaN 45.000000 NaN NaN 23.000000 151087.000000 NaN NaN}
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Gemini AI Insights

As an expert Data Analyst, I've reviewed the provided dataset summary. Here's a comprehensive analysis:

1. Dataset Overview

This dataset contains information on **200 employees** across **9 different attributes (columns)**. It's a clean dataset with **no missing values or duplicate rows**, which is excellent for immediate analysis.

The columns are:

- EmployeeID:** A unique numerical identifier for each employee.
- Name:** The name of the employee (unique for each record).
- Age:** The employee's age (numerical).
- Gender:** The employee's gender (categorical, either Male or Female).
- Department:** The department the employee belongs to (categorical, 6 unique departments).
- Experience:** The employee's years of experience (numerical).
- Salary:** The employee's annual salary (numerical).
- Performance:** The employee's performance rating (categorical, 3 unique levels).
- City:** The city where the employee is located (categorical, 5 unique cities).

2. Key Insights

- Dataset Size:** A moderate dataset of 200 employee records, sufficient for initial trends and patterns.
- Data Quality:** Excellent data quality with no missing values or duplicate entries, minimizing the need for immediate data cleaning.
- Employee Demographics:**
 - Age:** Employees range from 22 to 45 years old, with an average age of about 34.4 years. This

suggests a relatively young to mid-career workforce.

* **Gender:** There's a slight lean towards Female employees (106) compared to Male employees (94), indicating a fairly balanced gender distribution.

* **Experience:** Experience levels vary widely from 0 to 23 years, averaging around 10.3 years.

* **Organizational Structure:**

* **Departments:** The company has 6 distinct departments, with the **IT** department being the largest, accounting for 42 out of 200 employees. This might suggest IT is a core or rapidly growing area.

* **Cities:** Employees are spread across 5 different cities, with **Pune** being the most common location (49 employees).

* **Compensation and Performance:**

* **Salary:** Salaries range significantly from approximately \$26,500 to \$151,000, with an average salary of about \$83,370. The standard deviation (\$34,320) indicates a wide spread in salaries.

* **Performance:** There are 3 distinct performance levels. **"Excellent"** is the most frequent rating, given to 74 employees. This is a positive sign but also warrants further investigation into the distribution of performance ratings.

* **Unique Identifiers:** `EmployeeID` and `Name` columns are unique for each record, making them reliable identifiers.

3. Missing Value Analysis

The dataset has **zero missing values** across all 9 columns. This is a highly positive aspect, as it means the data is complete and ready for analysis without any imputation or removal strategies. This significantly reduces potential biases or complexities in the analysis phase.

4. Outlier Analysis

Based solely on the summary statistics:

* **Age:** The min (22) and max (45) values are reasonable for an employee population. The range and quartiles (25%: 29, 50%: 35, 75%: 40.25) appear well-distributed, suggesting no obvious age outliers.

* **Experience:** Experience ranges from 0 to 23 years. A 0-year experience employee is plausible (new hire). The max experience of 23 years is also reasonable. The distribution (25%: 4.75, 50%: 11, 75%: 17) seems normal, with no extreme jumps indicating outliers.

* **Salary:** While salaries have a wide range (\$26,501 to \$151,087) and a high standard deviation (\$34,320), the quartile values (25%: \$53,997, 50%: \$88,090, 75%: \$114,769) show a consistent spread. Without visualization (like a box plot), it's hard to definitively identify statistical outliers. However, the min and max values are within a plausible range for a diverse workforce, so they don't immediately appear as data entry errors. Further investigation with visualizations is recommended.

Conclusion: No strong evidence of extreme outliers from the summary statistics alone. The numerical data ranges appear sensible for an employee dataset.

5. Recommended Visualizations

To gain deeper insights, I recommend the following visualizations:

* **Univariate Distributions:**

* **Histograms:** For 'Age', 'Experience', and 'Salary' to visualize their distributions (e.g., normal, skewed).

* **Bar Charts/Count Plots:** For 'Gender', 'Department', 'Performance', and 'City' to show the count of employees in each category.

* **Bivariate Relationships:**

* **Scatter Plot:** 'Salary' vs. 'Experience' to explore if there's a correlation (likely positive).

* **Box Plots:**

* 'Salary' by 'Department' to compare salary ranges across departments.

* 'Salary' by 'Performance' to see if higher performance correlates with higher salaries.

* 'Salary' by 'Gender' to check for potential pay gaps.

* 'Salary' by 'City' to assess geographical pay differences.

* **Grouped Bar Charts:**

* Average 'Salary' by 'Department' and 'Performance'.

* 'Gender' distribution within each 'Department'.

* Average 'Experience' by 'Department'.

6. Business Recommendations

Based on the initial insights, here are some business recommendations:

* **Compensation Strategy Review:**

* **Fairness:** Investigate the wide salary range. Use visualizations (e.g., Salary vs. Experience, Salary by Department, Salary by Gender) to ensure fair compensation practices and identify any unexplained disparities.

* **Performance-based Pay:** Analyze if "Excellent" performers are indeed among the highest-paid, and if there's a clear salary progression with performance.

* **Talent Management & Retention:**

* **High Performers:** With 'Excellent' being the most frequent performance rating, understand what drives this and ensure these top performers are engaged and retained.

* **Career Progression:** Given the diverse experience levels (0-23 years), ensure clear career paths and development opportunities are available for employees at all stages.

* **Age Distribution:** The mid-career average age suggests a need for succession planning and leadership development programs.

* **Workforce Planning:**

* **Departmental Allocation:** The IT department is the largest. Assess if current staffing aligns with business needs and growth projections for all departments.

* **Geographic Strategy:** Pune has the most employees. Understand the strategic reasons for this concentration and its implications for office space, talent acquisition, and regional growth.

* **Diversity & Inclusion:**

* **Gender Balance:** While overall gender distribution is fairly balanced, investigate if there are significant imbalances within specific departments or at higher salary/performance tiers.

7. Future Analysis Suggestions

To further enrich the understanding of this employee dataset, I suggest the following analyses:

* **Correlation Analysis:** Quantify the relationships between numerical variables (e.g., Salary, Experience, Age) and explore how categorical variables (e.g., Department, Performance) influence them.

* **Regression Analysis:** Build a model to predict 'Salary' based on 'Experience', 'Age', 'Department', 'Performance', and 'City' to understand the key drivers of compensation.

* **Performance Driver Analysis:** Investigate what factors (e.g., Age, Experience, Department) are most associated with 'Excellent' performance.

- * **Departmental Deep Dive:** Analyze each department's specific characteristics (average age, experience, salary, performance distribution) to identify unique needs or challenges.
- * **Employee Segmentation:** Cluster employees into groups based on multiple attributes (e.g., high experience/high salary/excellent performance) to identify distinct employee profiles.
- * **Hypothesis Testing:** Formulate and test hypotheses, such as "Is there a statistically significant difference in average salary between male and female employees?" or "Does average experience vary significantly across different departments?".
- * **Sentiment Analysis (if qualitative data available):** If there were any survey comments or feedback, analyzing that could provide qualitative insights into employee satisfaction or concerns.
- * **Attrition Risk (if attrition data available):** If this dataset were augmented with attrition data, one could predict which employees are at higher risk of leaving the company.